

Quantification of the Improvement due to Transfer Learning in Segmentation of Neuromelanin-enriched Brain Structures

Tayebeh Bahador

Faculty of Electrical Engineering and Information Technology
Institute for Medical Engineering
Otto von Guericke University (OVGU)
Magdeburg, Germany
tayebeh.bahador71@gmail.com

Max Dünnwald

Faculty of Computer Science
Otto von Guericke University
Department of Neurology, Faculty of Medicine
Otto von Guericke University
Magdeburg, Germany
max.duennwald@med.ovgu.de

Abstract—The locus coeruleus (LC) and substantia nigra pars compacta (SNpc) are small brainstem regions involved in the early progression of Alzheimer’s disease (AD) and Parkinson’s disease (PD). Neuromelanin-sensitive magnetic resonance imaging (NM-MRI) enables their visualization and may support early detection of neurodegeneration. However, manual segmentation of these structures is difficult and time-consuming, and deep learning models often suffer from the limited availability of labeled data. Transfer learning (TL) may offer a possible solution by adapting pre-trained models to small datasets.

In this work, several TL strategies were evaluated using a 3D U-Net for LC and SNpc segmentation. TL improved performance compared to training from scratch when the dataset size was small. The best LC segmentation using TL achieved an average Dice Similarity Coefficient (DSC) of 53.16% ($\pm 22.8\%$), which is below typical intra-rater agreement but comparable to inter-rater agreement. For SNpc, the best DSC reached 74.08% ($\pm 12.15\%$), close to reported reproducibility levels. Training from scratch still achieved competitive DSC values with as few as 14 subjects. However, Intraclass Correlation Coefficient (ICC) analysis showed poor reproducibility of LC contrast ratios for TL and small datasets, indicating that DSC alone is not sufficient and should be complemented by ICCs.

Index Terms—Neuromelanin MRI, Segmentation, Deep Learning, Locus Coeruleus, Substantia Nigra pars compacta, TL, 3D U-Net.

I. INTRODUCTION

Neurodegenerative diseases such as Alzheimer’s disease (AD) and Parkinson’s disease (PD) are becoming more common as the population ages. Two small brainstem regions, the locus coeruleus (LC) and the substantia nigra pars compacta (SNpc), are experiencing cell loss in these diseases. Loss of neuromelanin-containing neurons in these regions decreases the dark-pigmented neuromelanin (NM), reflecting early neurodegenerative changes [1].

Neuromelanin-sensitive magnetic resonance imaging (NM-MRI) could help to visualise potential neurodegeneration. Since clinical symptoms appear in advanced stages of these diseases, early-stage biomarkers may help in early diagnosis and more efficient treatments [3]. Therefore, segmentation of

SNpc and LC could play an important role. Manual segmentations of these tiny structures are time-consuming and subjective to each rater. Hence, automated techniques are the subjects undergoing intense study.

In the last decade, convolutional neural networks (CNNs) and supervised learning have shown promising results in classification and segmentation of medical images that often outperform the classical approaches. The major issue in supervised learning is providing enough labeled datasets for training the models. Therefore, in this work different transfer learning methods are explored and evaluated in a convolutional neural network, namely the 3D U-Net.

In transfer learning methods, a pre-trained model is applied as training initialization, which may provide prior information leading to better segmentation performance. Because SNpc and LC are both part of the brain MR scans and are likely visualised using the same NM-sensitive MRI, we hypothesize that transfer learning from SNpc to LC or vice versa may improve the segmentation of these structures. To the best of our knowledge, there has been no study focusing on transfer learning for segmentation of LC and segmentation of SNpc in MR scans yet. Moreover, in similar works on TL for SNpc and LC, for the evaluation of models, ICC metric has not been considered into account. Therefore, in this work, the ICC metric for LC segmentation models has been calculated as well.

Therefore, the objectives of this work are:

- 1) Analyzing the performance of transfer learning on the state-of-the-art methods from LC to SNpc and vice versa, to evaluate whether the network can benefit from this prior information.
- 2) Comparison of the performance of different transfer learning techniques.
- 3) Comparing different TL setups and training sizes to find the best transfer learning approach for segmentation of LC and SNpc.
- 4) Finding the number of samples in the dataset, on which transfer learning methods would have a promising

performance.

II. RELATED WORK

A. Segmentation Approaches in Brain MRI

Brain MRI segmentation has been widely studied using both classical image processing and deep learning methods [4]. Manual segmentation remains the gold standard but is time-consuming and prone to intra- and inter-rater variability [5]. Tools such as ITK SNAP [6] support manual annotation but are not scalable for large datasets.

Early automated methods, including thresholding, region growing, and atlas-based registration [7]–[9], often struggle with noise and low tissue contrast. Deformable surface models [10] improved boundary detection but remained sensitive to initialization and image artifacts.

B. Deep Learning and Neuromelanin MRI

Convolutional neural networks (CNNs) have become the dominant approach for medical image segmentation [11], [12]. The 3D U-Net architecture enables effective feature extraction and accurate voxel-wise prediction, and its variants such as U-Net++ and Attention U-Net enhance feature fusion and focus [13], [14].

In NM-MRI, Krupicka et al. [15] and Le Berre et al. [16] applied a U-Net for substantia nigra segmentation, achieving Dice scores close to inter-rater agreement. Dünnwald et al. [17] extended this work to the locus coeruleus (LC), showing improved reproducibility over manual segmentation.

C. Transfer Learning in Medical Imaging

Transfer learning (TL) helps mitigate data scarcity by reusing pre-trained CNNs and fine-tuning selected layers [18], [19]. Studies have shown TL to improve segmentation in various modalities and diseases [20]–[22].

However, no previous work has systematically evaluated TL for LC and SNpc segmentation in NM-MRI. This study addresses this gap by comparing multiple TL strategies for these small, clinically relevant brainstem structures.

III. MATERIALS AND METHODS

This study investigates transfer learning approaches using a 3D-U-Net architecture for automated segmentation of the LC and SNpc in NM-MRI. Choosing this approach was inspired by the work [16], which used a U-Net architecture for segmentation of SNpc in NM-MRI scans, and the work by Dünnwald et al. [17], which applied a 3D-U-Net architecture for LC segmentation. The latter study used the same data that was used in this study and acquired around 70% DSC value, which outperformed the inter-rater agreement.

A scheme of the whole network can be seen in figure 1. Similar to the work [17], the network includes three down-sampling blocks, one bottleneck block, and three up-sampling blocks, and one convolutional layer followed by a sigmoid function as the last layer. The reason for using a sigmoid function was to get the outputs between zero and one. Each block has two convolutional layers with $3 \times 3 \times 3$ filters,

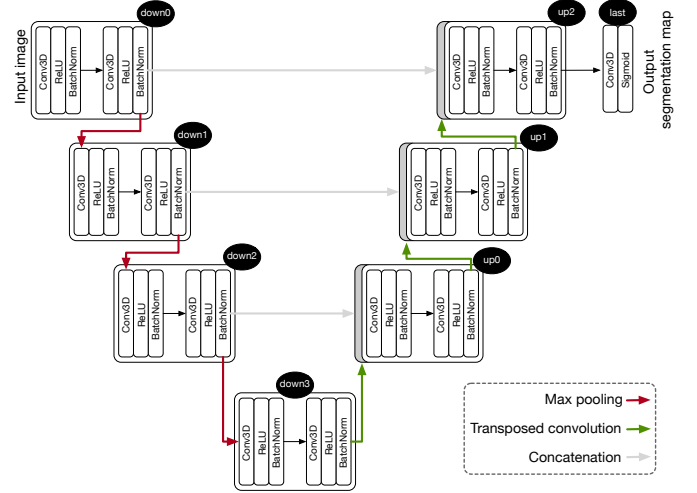


Fig. 1. A scheme of the applied network in this thesis. On top of each block, its label is depicted in black.

followed by ReLU activation and batch normalization [23] to normalize the output of each layer.

Down-sampling is performed by $2 \times 2 \times 2$ max pooling to reduce the feature map dimensions and make the model more robust to spatial variations. Transposed convolution was used in the up-sampling path. Moreover, the padding technique was applied. Therefore, input and output have the same sizes [24].

A. Dataset

The dataset used in this work is provided by the German Center for Neurodegenerative Diseases [2]. The dataset comprises high-resolution isotropic T1-weighted FLASH MR scans, acquired at 3T. It includes 57 old (19 male, 38 female) and 25 young (13 male, 12 female) healthy subjects. Isotropic voxel size was 0.75mm. The images matrix size was $320 \times 320 \times 192$. Moreover, the images were convolved with a sinc filter to be upsampled. Therefore, the final matrix size was $639 \times 639 \times 383$ and the isotropic voxel size was 0.375 mm. Duration of a scan was 13:50 minutes. Moreover, the manual segmentations for SNpc and LC as ground truth for training, as well as LC reference region masks for further analysis of contrast ratio, were created by a trained expert.

B. Training

Figure 2 presents an example of the input image along with the ground truth masks of SNpc and LC.

The loss function for training was the Dice Similarity Coefficient (DSC) [17]:

$$DSC = \frac{2|X \cap Y|}{|X| + |Y|} \quad (1)$$

where X and Y denote the predicted and ground-truth masks, respectively.

Adaptive Moment Estimation (Adam) [25] was used as stochastic optimizer, with a learning rate of 0.001 and a random batch size of 32, due to the memory limitation of the Graphics Processing Unit (GPU) that did not allow for a larger

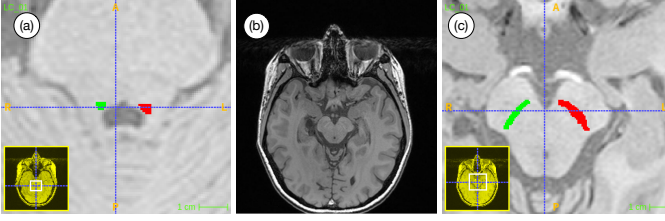


Fig. 2. An example of the input and SNpc and LC ground truth masks. (a) Axial view of LC manual segmentation as training ground truth. (b) A slice of one of the volumes used as training input. (c) Axial view of SNpc manual segmentation as training ground truth.

batch size. This optimization method just uses first-order and second-order derivatives. Therefore, it may need less memory consumption and could be a suitable choice for problems with higher number of parameters. This optimizer sets a different learning rate for each parameter. First, one model was trained from scratch on 81 number of SNpc masks and one model on 81 number of LC masks. The input of these trainings were the whole image volume. The final trained weights of these two models were then used as the initialization in training the models with transfer learning techniques.

Each model was trained for 2500 epochs. After every 10 epochs, weights were updated using the DSC metric to improve the model performance. Random 3D patches of size 64^3 voxels were extracted from the input volumes [17], [26]–[28], with a 50% probability of including the target structures to balance positive and negative samples and avoid a negative bias. Using patches brings up some benefits. First, as a method for data augmentation, it leads to having more number of data for training. Second, it is computationally less expensive in comparison with feeding the network with the entire image. Data augmentation included random rotations along all three axes with 50% probability [11]. All models were trained using 5-fold cross-validation [16] so that the performance analysis can be more relevant. So the model was trained on 4 folds and validated on the last remaining fold. The dataset was randomly split into training, validation, and test sets, with 27% reserved for testing. The model weights with the lowest validation loss were saved as the best model.

C. Transfer Learning Setup

In this study, two TL strategies were compared against training from scratch. The experiments were performed on datasets of different sizes to evaluate how TL affects segmentation performance in low-data conditions.

Inspired by previous transfer learning studies, two transfer learning (TL) strategies were evaluated using a 3D U-Net with three encoder blocks, one bottleneck, and three decoder blocks. In the first approach, selected layers were gradually unfrozen from the output layer toward the encoder, while frozen layers had their learning rate set to zero. In the second approach, instead of zero, the learning rate of the “frozen” layers was set to a tenfold smaller value (0.0001) compared to the fine-tuned layers (0.001), following [17]. Each configuration was

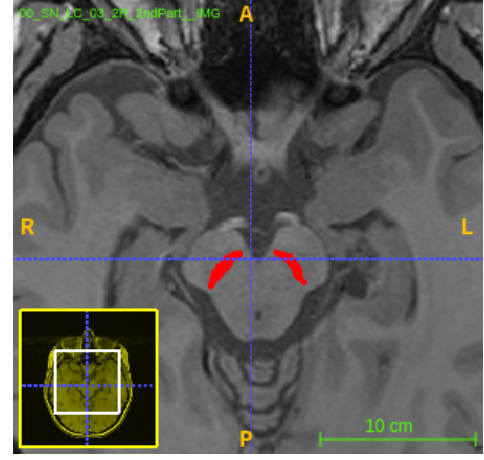


Fig. 3. An example of network output mask for SNpc overlaid on the input image using ITK-SNAP software.

tested by fine-tuning increasing portions of the network until the entire model was trainable. Both TL strategies and training-from-scratch baselines were trained on the full dataset (81 subjects) and then repeated with reduced subsets of 62, 42, and 22 subjects to analyze the effect of dataset size on segmentation performance.

We used pretrained weights from trained models on 81 labeled LC to initialize SNpc training and vice versa (cross-region TL). For the sake of a valid comparison, all other hyper-parameters were also kept constant in all the models.

To ensure reproducibility, random seeds were fixed for data splits and weight initialization. The training and testing code were executed on the same system to avoid hardware-related performance variations.

D. Testing and Evaluation

An example of a network output mask overlaid on the input image is shown in figure 3. Performance was measured using DSC for both LC and SNpc. During inference, a sliding window with patch size 128^3 and overlapping by half of the patch size was applied to each test volume [17]. The resulting output patches were merged and thresholded at 0.5 to form the final segmentation mask.

For LC, the maximum and median contrast ratios relative to the pontine tegmentum reference region [3] were computed, and the intraclass correlation coefficient (ICC) between predicted and manual masks was used to assess reproducibility. ICC was not calculated for SNpc due to the lack of reference region masks. Considering ICC helps for evaluating the reproducibility of the contrast ratio.

IV. RESULTS AND DISCUSSION

A. Evaluation protocol

We report DSC values for the combined left and right structures for both LC and SNpc segmentation, for each model trained with different dataset sizes. All scores are averaged over the test set with 5-fold cross-validation unless stated otherwise.

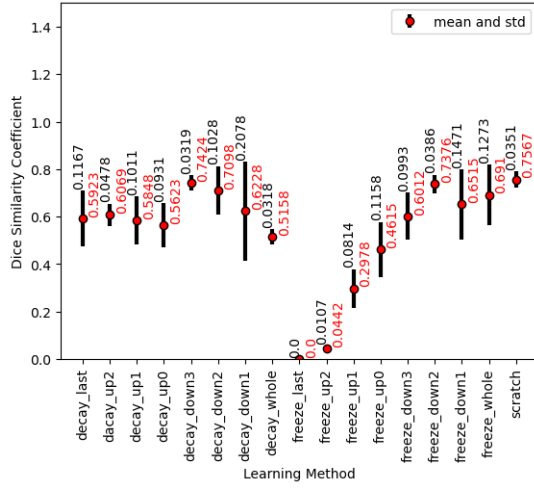


Fig. 4. The mean and STD of the DSC values in different learning methods for segmentation of the combined SNpc using 22 target dataset size.

As a practical reference, prior work [29] reports manual reproducibility around $DSC \approx 0.80 \pm 0.03$ for SNpc and $\approx 0.63 \pm 0.07$ for LC, with $ICC \approx 0.94$ (SNpc) and ≈ 0.96 (LC) for contrast ratios. We use these ranges to interpret our results. However, the mentioned values would be taken with a grain of salt, because their experiments were also limited to only a few number of raters or a few number of scans.

B. Main quantitative findings

The results could be seen in the bar charts from the figure 4 to the figure 11. The best learning method could be a method with a higher mean DSC value and lower STD across the folds.

The annotations used in the bar charts are illustrated in Figure 1. "Last" denotes fine-tuning only the final 3D convolutional layer. "Up2" refers to fine-tuning the final 3D convolutional layer together with the last block of the upsampling path. Similarly, "Up1", "Up0", "Down3", "Down2", and "Down1" indicate fine-tuning from the corresponding block onward until the end of the network. "Whole" denotes fine-tuning the entire network.

The freeze-last, freeze-up2, freeze-up1, and freeze-up0 strategies consistently showed poor performance. This suggests that fine-tuning only the later layers is insufficient and that adapting some blocks in the encoder path may be necessary to achieve acceptable segmentation performance. Furthermore, learning rate decay strategies generally outperformed the corresponding frozen strategies.

C. Effect of dataset size

To enable valid comparisons across different dataset sizes, additional experiments were conducted using 14, 7, and 5 training samples with a fixed test set derived from the 81-sample setting, ensuring that the test data remained unseen by all models. The 14-, 7-, and 5- sample datasets were obtained from the training-validation sets of the 81-, 14-, and

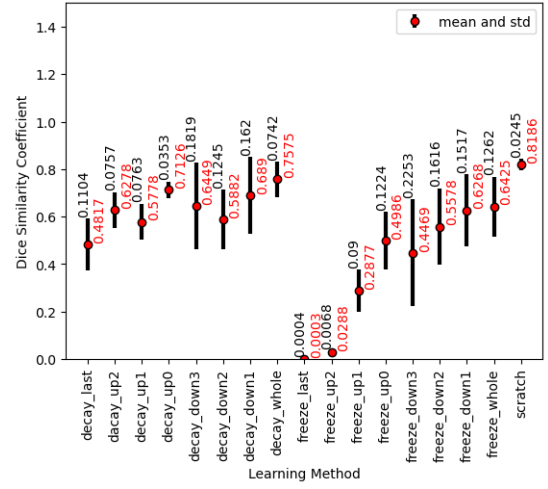


Fig. 5. The mean and STD of the DSC values in different learning methods for segmentation of the combined SNpc using 42 target dataset size.

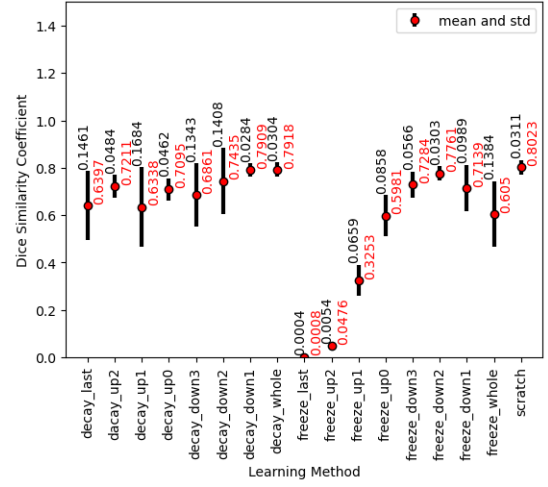


Fig. 6. The mean and STD of the DSC values in different learning methods for segmentation of the combined SNpc using 62 target dataset size.

7- sample settings, respectively, and 5-fold cross-validation was performed to improve the robustness of the evaluation. These smaller dataset sizes were selected based on previous findings to test whether further data reduction would cause a larger DSC drop or identify conditions where transfer learning could outperform training from scratch. These models were trained with 10000 numbers of epochs to let the networks converge.

A comparison of DSC values of each model for each dataset size can be found in the charts from the figure 12 to the figure 17.

For these models, testing was performed using a 128^3 patch extracted around the center of mass of the target structure instead of the sliding-window approach, resulting in faster inference and reduced false positives by focusing on the region of interest. Additionally, post-processing was applied to retain only connected components containing at least 50 voxels, and

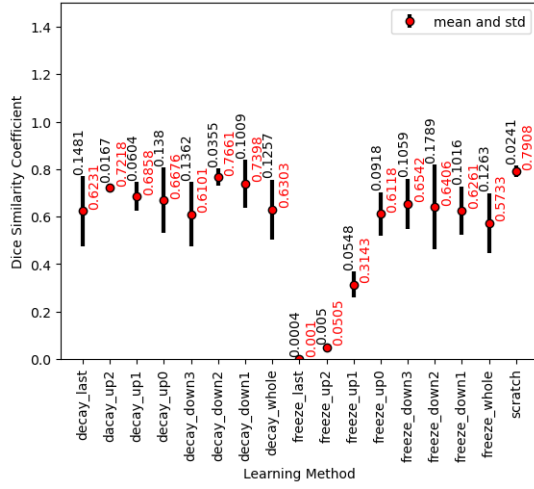


Fig. 7. The mean and STD of the DSC values in different learning methods for segmentation of the combined SNpc using 81 target dataset size.

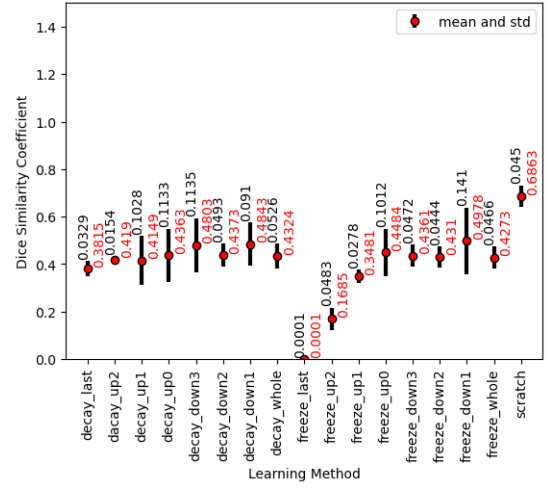


Fig. 9. The mean and STD of the DSC values in different learning methods for segmentation of the combined LC using 42 target dataset size.

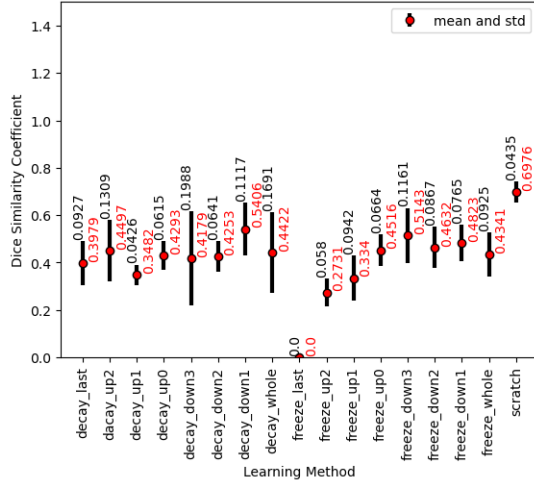


Fig. 8. The mean and STD of the DSC values in different learning methods for segmentation of the combined LC using 22 target dataset size.

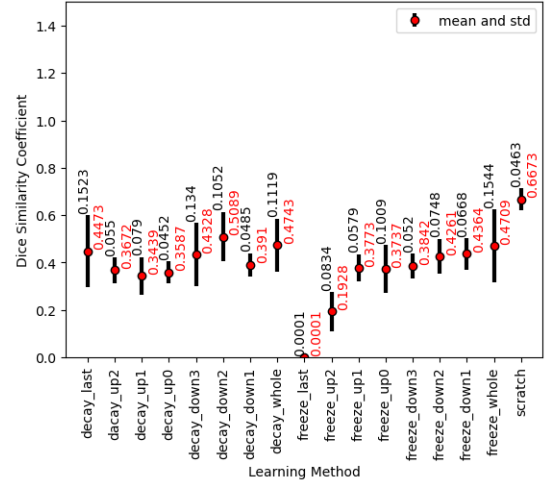


Fig. 10. The mean and STD of the DSC values in different learning methods for segmentation of the combined LC using 62 target dataset size.

the DSC values were subsequently calculated from the post-processed segmentation masks.

As the figures 12 and 15 depict, for smaller dataset size of 5, transfer learning approaches helped and could achieve higher DSC values in comparison with the training from scratch. With 5 number of dataset, the DSC value for the most promising transfer learning method in segmentation of LC was 52.01% ($\pm 14.74\%$), figure 12, and for SNpc was 73.54% ($\pm 11.92\%$), figure 15. The achieved DSC value for LC is relatively low in comparison with the intra-rater agreement ranged from 0.65 to 0.74. However, the average DSC is more comparable to the inter-rater agreement ranged from 0.54 to 0.64. The achieved DSC value for SNpc could be comparable to the SNpc reproducibility, which is approximately 80% (± 0.03).

In General, across all the models and dataset sizes, the best LC segmentation with TL reached **54.73% $\pm$ 11.17** DSC, figure 14, and the best SNpc result reached **79.18% $\pm$ 3.04**

DSC, figure 6. While LC DSC is again below typical intra-rater agreement, it is within inter-rater ranges [?]. SNpc DSC is almost equal to reported reproducibility.

These results could show that transfer learning techniques could achieve a better performance in the SNpc segmentation in comparison with the LC segmentation.

The training methods with higher DSC values across different number of dataset sizes for the segmentation of left and right LC was compared and shown in figure 18 and 19. The DSC mean of all folds in each method and each data set size were compared. Each DSC value had a considerable STD, which is not depicted in the figures. The DSC mean of the training from scratch in both right and left LC was reduced by the reduction in number of dataset. However, this reduction is only around 5% until 14 number of dataset. In the left LC there is a same reduction in decay down1. However, other transfer learning methods, freeze down1, freeze down2

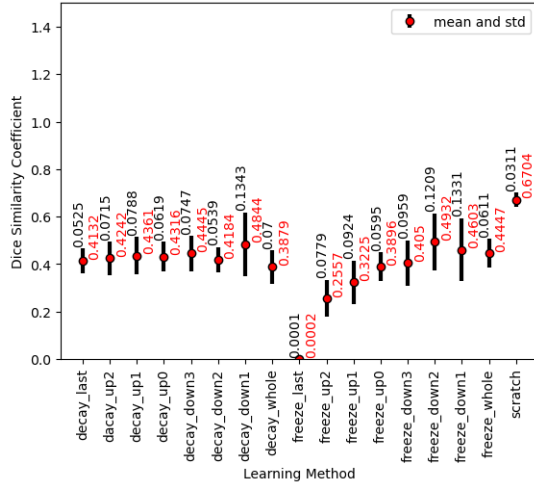


Fig. 11. The mean and STD of the DSC values in different learning methods for segmentation of the combined LC using 81 target dataset size.

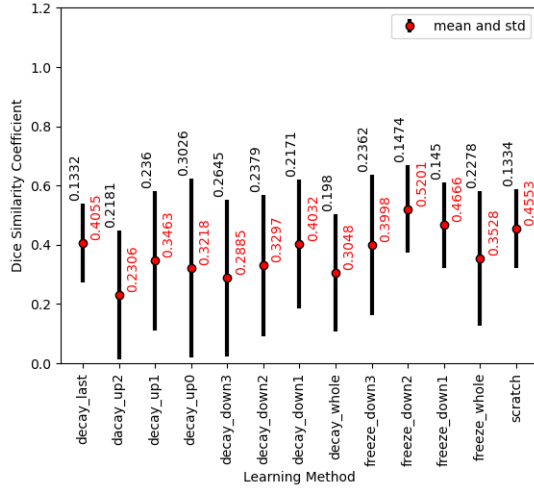


Fig. 12. The mean and STD of the DSC values in different learning methods for segmentation of the combined LC using 5 target dataset size.

and decay whole were all achieving higher DSC while the number of dataset size was reducing to 14. In the right LC, among the transfer learning methods, decay down1 achieved a higher mean DSC until 14 dataset and freeze down1 had a huge reduction in the mean DSC until 14 dataset. Therefore, the pattern in changes for decay down1 and freeze down1 for left and right LC were in the opposite directions. This could be due to the high STD of the DSC values of these methods in each dataset size, which is not depicted in the figure.

The same comparison was performed for the left and right SNpc, figures 20 and 21. For the right SNpc trainings from scratch, decay down2, and freeze down2 TL approaches, had approximately similar DSC values. Moreover, they have a reduction less than 5% until the reduction of dataset size to 14. On the other hand, the mean DSC values of the TL methods for left SNpc changed differently. Decay down1 and decay whole had a reduction in the DSC, but decay last, freeze down2 and

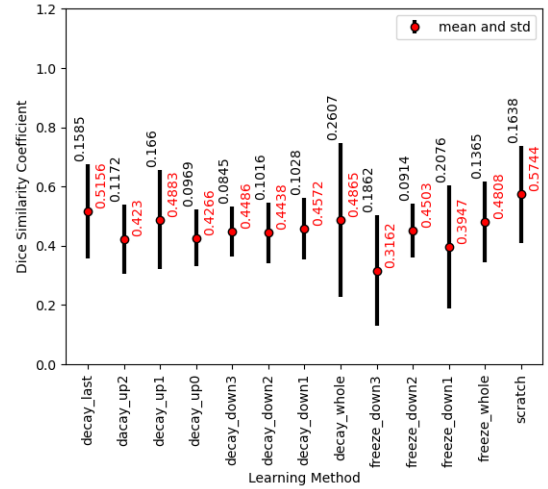


Fig. 13. The mean and STD of the DSC values in different learning methods for segmentation of the combined LC using 7 target dataset size.

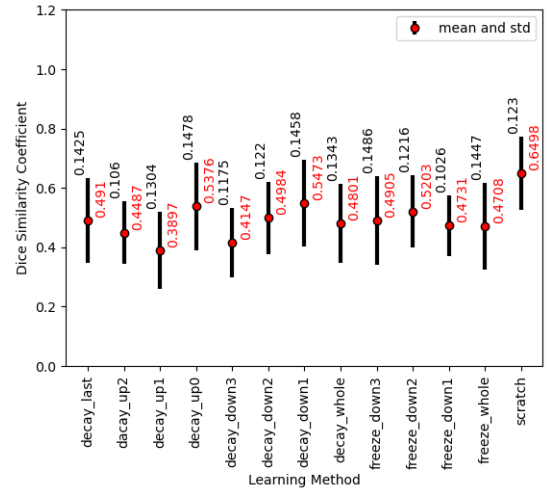


Fig. 14. The mean and STD of the DSC values in different learning methods for segmentation of the combined LC using 14 target dataset size.

freeze whole DSC values increased by the reduction in dataset sizes. The mean DSC of the trainings from scratch reduced by less than 5% until 7 number of datasets. The differences in the performances of SNpc segmentation methods could be due to the reason that some of the models could not converge in 10000 epochs.

D. Transfer direction and fine-tuning depth

We tested LC→SNpc and SNpc→LC. TL toward SNpc gave clearer gains than toward LC. Freezing many layers (learning rate = 0) performed poorly. Using a smaller learning rate in the “frozen” part (differential LR) was more stable. Gradually unfreezing deeper blocks helped up to a point, and fully fine-tuning worked best when more data were available. In short: *reduce the LR rather than hard-freeze, and fine-tune more blocks when you can.*

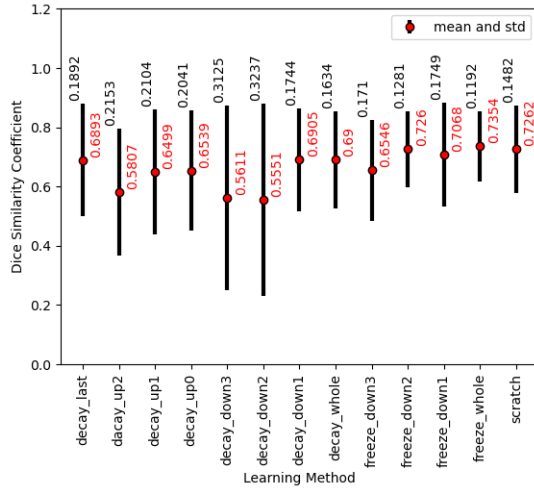


Fig. 15. The mean and STD of the DSC values in different learning methods for segmentation of the combined SNpc using 5 target dataset size.

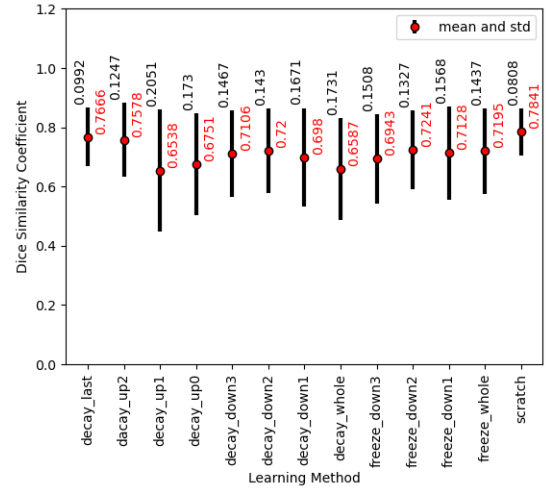


Fig. 17. The mean and STD of the DSC values in different learning methods for segmentation of the combined SNpc using 14 target dataset size.

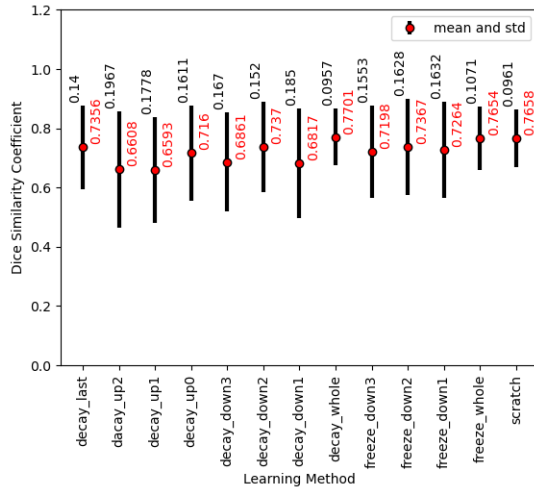


Fig. 16. The mean and STD of the DSC values in different learning methods for segmentation of the combined SNpc using 7 target dataset size.

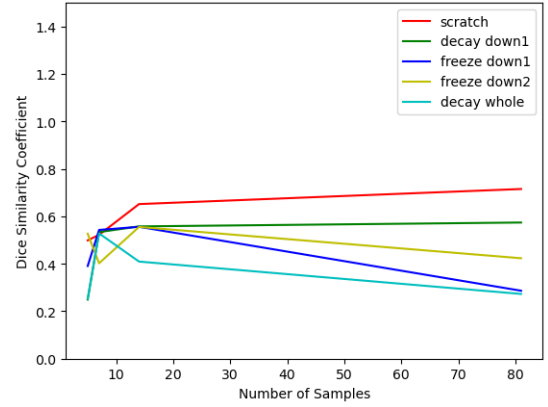


Fig. 18. The DSC values of promising learning methods for segmentation of the left LC across different number of datasets.

E. Cross-structure generalization without TL

Models trained from scratch on LC, for all dataset sizes 22, 42, 62, and 81, were evaluated directly on SNpc (and vice versa) and produced $DSC \approx 0$. Even though both structures are small brainstem region, a model trained on one cannot simply be reused for the other without adaptation. TL is required.

F. Impact of Random Data Splits on Model Performance

To assess the effect of dataset randomness on model performance, the experiments for LC segmentation using 7 training samples were repeated with a different random seed, figure 22. The results showed that training from scratch generally remained superior to transfer learning, while the best-performing transfer learning strategy varied across seeds, indicating that model performance was influenced by the random data split.

G. ICC findings and metric considerations

For LC, ICC between contrast ratios from predicted versus manual masks was close to zero for TL in small-data settings, and it dropped as data decreased for models trained from scratch. This gap between DSC and ICC suggests that DSC alone is not enough for very small, low-contrast targets. We recommend reporting DSC together with ICC, and adding boundary-sensitive metrics (e.g., surface distances) in future work.

H. Qualitative observations

Three points stand out:

- 1) For LC segmentation, "decay down1" most frequently achieved the highest DSC among the transfer learning methods, whereas "decay whole" was the most successful strategy for SNpc segmentation. These findings suggest that reducing the learning rate is more effective than completely freezing layers and that fine-tuning deeper portions of the network, including parts

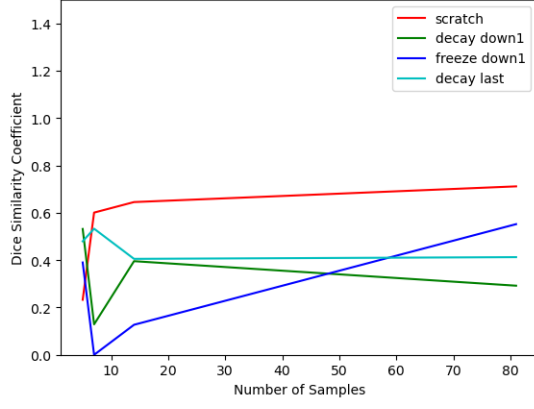


Fig. 19. The DSC values of promising learning methods for segmentation of the right LC across different number of datasets.

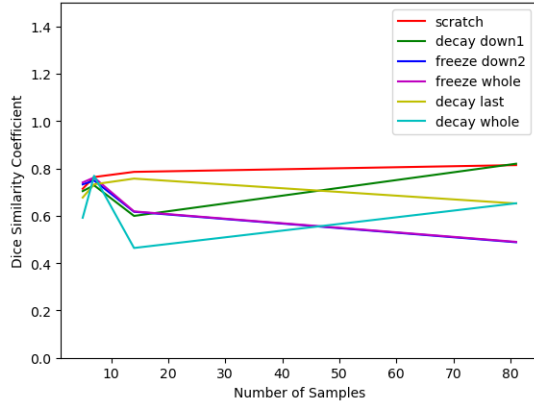


Fig. 20. The DSC values of promising learning methods for segmentation of the left SNpc across different number of datasets.

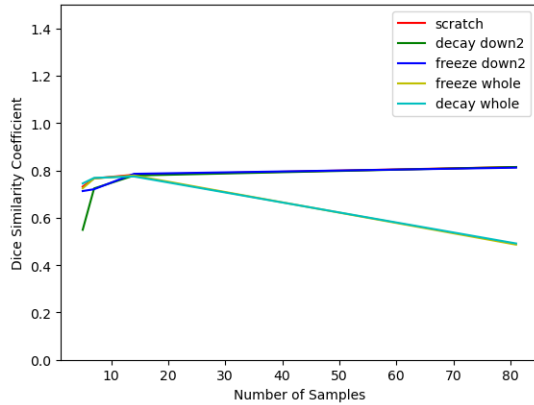


Fig. 21. The DSC values of promising learning methods for segmentation of the right SNpc across different number of datasets.

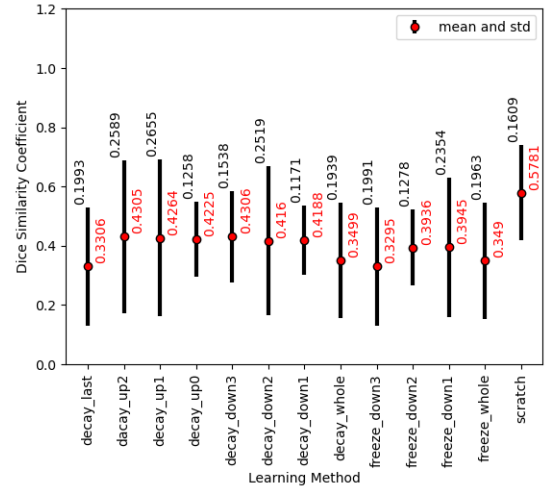


Fig. 22. The mean and STD of the DSC values in different learning methods for segmentation of the combined LC on 7 number of dataset using different seed.

of the downsampling path, may be necessary to adapt to anatomical differences between LC and SNpc.

- 2) 14 number of dataset could be sufficient for the segmentation of LC and SNpc from scratch.
- 3) TL is useful when data are scarce. With very small sets of 7 or 5 subjects, TL outperformed or matched training from scratch for both targets, especially for SNpc.
- 4) With more data, scratch can catch up. At 62/82 subjects, training from scratch matched or slightly surpassed TL.
- 5) SNpc achieve better results than LC through TL. In the models trained with 14 subjects, the most promising transfer learning methods for LC segmentation showed about a 10% lower DSC compared to training from scratch. In contrast, the maximum decrease in DSC for SNpc segmentation was only around 3%. Higher and more stable DSC for SNpc reflects its larger size and clearer appearance in NM-MRI, and that the segmentation of SNpc could be an easier task to learn in comparison with LC.

I. Summary

TL improves LC and SNpc segmentation when the dataset is small, with a stronger effect for SNpc. For larger datasets, training from scratch is competitive. For LC, ICC highlights that DSC alone may overestimate quality; mixed metrics are needed for a reliable evaluation.

V. CONCLUSION AND FUTURE WORK

This work investigated transfer learning (TL) methods for segmenting the locus coeruleus (LC) and substantia nigra pars compacta (SNpc) in neuromelanin-sensitive MRI. The goal was to address the lack of large annotated datasets, which often limits CNN performance in small-structure segmentation.

Our experiments showed that TL improves performance when the training data are very limited. With only five subjects, the best TL approach reached an average Dice score of

53.16%±22.8 for LC and 74.08%±12.15 for SNpc. While LC results remain below typical intra-rater agreement (0.65–0.74), they are close to inter-rater values (0.54–0.64). The SNpc results are near the reproducibility reported in the literature (~ 80%). These findings indicate that SNpc segmentation is more stable and that TL is especially effective for this structure.

Across different TL strategies, the “decay down1” method performed best for LC and “decay whole” for SNpc. Reducing the learning rate in deeper layers (rather than freezing them) generally led to higher Dice scores. Fine-tuning more blocks, including parts of the encoder, was beneficial when adapting between LC and SNpc domains. However, results were somewhat sensitive to random initialization.

When around 14 data were available, training from scratch was still more competitive, showing only a small drop (around 5%) compared to larger datasets, for both SNpc and LC. ICC analysis suggested that TL may not preserve contrast reproducibility in LC segmentation, where ICC values dropped well below the expected 0.96. This implies that Dice alone may not fully capture segmentation reliability for small, low-contrast structures.

Future Work

Several directions could extend this study:

- Our dataset is modest and from a single site and sequence. Labels come from one expert. We did not study domain shifts across scanners or protocols. Future work will test semi-/self-supervised pretraining on NM-MRI, add boundary-aware losses and ROI cropping for LC, report surface distances alongside DSC/ICC, and evaluate on multi-site data including patient scans.
- Provide the masks of reference region for SNpc to compute ICC values to assess whether TL also improves contrast ratio reproducibility in this structure.
- Explore additional dataset sizes between 14 and 81 subjects using a fixed test set, to better understand how performance scales with data.
- Evaluate TL from non-medical pretraining sources such as ImageNet, following [?], to test domain transfer effects.
- Investigate data-efficient methods such as Deep Image Prior [?], which can exploit network structure as a prior even with a single image.

Overall, our results suggest that TL can meaningfully improve segmentation performance in small datasets, particularly for the SNpc. However, further work is needed to ensure reproducibility and to explore complementary strategies for LC, where the task remains more challenging.

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